Topic B: proof and source scope

Human Compression project

This private author edition reorganizes four existing accepted contracts. It admits no additional source or question result. The common paper and all four applications form the proposed scientific volume. The original source statements, their prior acceptance, and the precise new proof spans are recorded in `source_contract_map.json` and `proof_map.json`. The earlier combined manuscript is a migration record, not an unpriced mathematical premise of this edition.

Paper	Source	Fixed principal contract
B1	Buekenhout–Cara (1998)	Theorem 1, all four clauses
B2	Cameron–Cara (2002)	Eight stated results, $n \geq 7$
B3	Archer–Cara–Krempa (2005)	Eleven stated clauses, with scope below
B4	Garcia-Colin–Leemans (2021)	Seven clauses in the revised edition

1 What is shared and what is retained

The common paper proves the complete elementary coset interface, the residual-maximality induction and its geometric adapter, the positive weighted-forest construction, and the two-block overgroup interval. The forest includes both subgroup intersections and a common representative for every partial flag. It is used with unit weights, one contracted edge of weight two, and equal leaf weights. These are actual applications to specified subgroups, not a diagrammatic analogy. The Frattini ladder, the normal-quotient bound, and graph reconstruction are local proofs in the relevant papers. No application needs the proof of another application.

B1 retains the 1997 classification and uniqueness theorem of Buekenhout, Cara and Dehon in the exact form recalled by the 1998 source. B2 retains Whiston's independent-family bound and the separately stated omission consequence, including the latter's dependence on the classification of finite simple groups. Cayley's formula and the innerness of automorphisms of S_n for $n \geq 7$ are retained classical facts. B3 retains the elementary-abelian Frattini quotient of a finite p-group; its infinite maximal-overgroup argument uses Zorn's lemma. B4 proves the required Kneser filtration, eigenvalue bounds and equality eigenspace; only elementary real symmetric spectral theory and linear algebra are used without proof. These external costs remain explicit and unmeasured.

2 Source editions and qualifications

B1 uses the complete available authored browser text. Its mathematical formula and diagram extraction is imperfect, and the original PDF has not been acquired. Its source PDF cost is unknown. B1 remains part of the proposed four-application volume; it is not silently removed from the denominator or assigned an invented numerator. The source phrase "this is the case" is interpreted precisely as the criterion for failure of PRI, rather than for QPRI alone.

B2 uses the twelve-page author manuscript. Its extremal classification assumes $n \geq 7$ and retains three distinct counting conventions. Numerical group bounds are asserted for finite groups. Small-degree discussions and unnumbered general direct-product questions are outside this contract.

B3 uses the nine-page published source. The unrestricted infinite converse of its generating-set Theorem 2.1 is false; the paper includes both the relative finite-generation correction and its explicit counterexample. The infinite geometric containment theorem remains fully proved for finite-rank firm RWPRI geometries. The p-group classification requires finite groups and applies to every prime. Its count is of indexed subgroup families, not automorphism orbits. The residue Frattini condition is necessary, not sufficient. The quoted Gaschütz theorem and the older primitive-group reduction are outside the target clauses and are not used in their proofs.

B4 uses the ten-page institutional revised manuscript, whose title and numbering differ from other editions. The source map binds Theorem 1.2 and Corollaries 1.3, 1.4, 4.8 and 5.1. The three retained corrections are the panel order $\binom{n-k}{k} - 1$, the thin case $k = 1, n = 3$, and the full locally Desargues graph group $S_7 \times S_3$. The latter includes an uncolored type-reconstruction proof. No identity with an unacquired publisher edition is presumed.

3 What Lean verifies

The file `formal/ResidualGeneration.lean` works with actual subgroups of an arbitrary group. For a finite set S of unfixed types, it defines the intersection of the stabilizers outside S and the subgroup generated by the common stabilizer and the chosen elements in S. Off-diagonal membership, exclusion from one's own stabilizer, and a maximal next intersection imply equality of those subgroups by strong induction. The six reported declarations certify containment, that equality, the intersection formula, its generated-subgroup counterpart, irredundancy, and generation of the whole group for a finite type set. The desired equality is a conclusion, not a hypothesis.

The geometric panel choices and residual-maximality bridge are written proofs in UB.Coset, UB.Flags and UB.Generation. The forest construction, Whiston and B97 inputs, Frattini theorem, spectral argument, and four source applications are not formalized by this module. The checker freshly compiles the actual module against pinned Lean 4.32.2 and mathlib, reports only the standard axioms, rejects three missing-hypothesis proof edits, and checks three literal finite countermodels plus an altered model control. These examples establish the necessity of the stated hypotheses in the displayed cases; they do not replace any uniform proof.

4 Replay and accounting boundary

From the packet directory run

```
python3 -O check.py --out /fresh/absolute/path
```

The `--lean` and `--packages` options relocate the pinned toolchain and dependencies. With an interpreter providing `pypdf`, `--render` performs two XeLaTeX passes for all five scientific papers and this reader, checks diagnostics, and compares every page's extracted text with its authoritative PDF. The optional `--verify-primary --sources-root /private/study` checks the private source and prior-review bindings. Primary texts and images are not distributed in the packet. Cached dependency objects are part of the stated Lean trust boundary; no claim is made that mathlib was rebuilt.

No mathematical computation is required by the written uniform proofs. The portable checker is validation infrastructure, whose cost is recorded separately from the scientific papers if measured later. The complete common paper is paid within this topic even where arguments overlap another topic. This packet makes no numerical factor, full-visible-text comparison, or claim of a sawtooth pattern. The current mathematical edition remains an author candidate pending independent review.